VITAL Training School - Delft 18-21/03/2025
Modelling Workshop

Exercise: Analysis of arterial pulse data

Load the PWDB data and subsequently extract time, pressure, and flow vectors for one of the virtual subjects. The extracted data correspond to pressure and flow at the aortic root.

(i) Calculate input impedance in the frequency domain. The input impedance (Z_{in}) is calculated with the function 'calculate_input_impedance'. The Z_{in} modulus and phase of the first 11 harmonics will be displayed on the screen. The characteristic impedance (Z_c) can be calculated from 'calculate_characteristic_impedance'.

(ii) Estimate compliance using the pulse pressure method. The function 'calculate_pulse_pressure' will yield the response of the 2-element WK to the true flow wave, using the calculated value for total peripheral resistance along with a compliance estimate <u>that you have to provide</u>. Change the compliance value until there is a satisfactory agreement between true and predicted pulse.

(iii) Fit the parameters of the 3-element WK model.

(iv) Assuming blood density ρ = 1,060 kg/m^3, calculate the pulse wave velocity (PWV)? <u>Note</u>: Z_c can be approximated as ρ*PWV/A). Use calculate_pwv.